Resume

(As of 2025/06/23)

Name	Han	Jixin	Gender	**Male**
Date of birth	December 21, 1992	Nationality	China	
e-mail	kalfazed@gmail.com			

Education Background

Sep. 2010	**Entry**	Depart of Media and communication, Beijing Language University (Transferred to the Faculty of Information Sciences, Beijing University of Languages)
Mar. 2011	**Transfer**	
Apr. 2011	**Entry**	Department of Computer Science, Faculty of Information Science, Beijing Language University
Aug. 2014	**Graduate**	
Sep. 2014	**Entry**	Department of Computer Science and Engineering, Graduate School of Fundamental Science and Technology, Tokyo University. (Transferred to Waseda University later)
Aug. 2015	**Transfer**	
Sep. 2015	**Entry**	Department of Information Science and Technology (Skipped Grade), Graduate School of Fundamental Science and Technology, Master Course, Waseda University
Mar. 2017	**Graduate**	
Apr. 2017	**Entry**	Department of Information Science and Technology, Graduate School of Fundamental Science and Technology, Doctoral Course, Waseda University
Mar. 2022	**Graduate**	
Degree	**Ph.D. (Engineering)**	**Date of receipt** **March 15, 2022**

career

July 2024 ~ Present	**TIER-IV, Senior AI Engineer and Researcher, Future solution** At TIER-IV, I am mainly responsible for the following tasks: model prototyping, model deployment and integration, auto-labeling, and MLOps. In model prototyping, I have developed multi-task models leveraging techniques such as multi-modal sensor fusion, active learning, uncertainty estimation, and model ensembling to generate high-quality auto-labels. These pseudo-labels are then used to train lightweight models optimized for deployment on edge devices. Besides auto-labeling, I also deploy and integrate trained models using deep learning compilers such as TensorRT and HailoRT. In addition to utilizing built-in techniques provided by these compilers (e.g., quantization, scheduling, etc.), I have also developed custom toolkits for model bottleneck analysis, sensitivity analysis, layer-wise pruning, and more. Lastly, I use Kubernetes and Kubeflow to build MLOps pipelines that enable CI/CD/CT for data generation, data cleaning, training, deployment, and evaluation. Tasks are allocated resources dynamically within the GPU cluster based on their computational needs. Skills: C/C++; CUDA, TensorRT; PyTorch, k8s, kubeflow, ROS2
January 2024 ~ June 2024	**T2 Auto, Tech Lead, Perception model deployment and integration team** T2 auto is a startup company, where most of the software prototype is under development. When I was in T2 auto, I belong to the perception team. My job was to develop a BEVFusion model and deploy the model into NVIDIA architecture. The BEVFusion model uses LiDAR and camera as input sensors. I tried several different tactics to reach high precision and fast inference speed. For example, develop customized BEVPool methods to accelerate the projection from camera view to BEV grid, or use deformable attention as a substitute of BEV Pool. Besides, in the aspect of deployment, I use TensorRT and CUDA to optimize model. I also develop some benchmark tools for both training model and deployed model using C++. For model integration, I also develop synchronization algorithm across difference sensors. Skills: C/C++; CUDA & cuDNN & TensorRT; PyTorch, Apollo, protobuf
April 2023	**Waseda University, Graduate School of Information Science and Technology, Visiting Researcher**

～Present	**and Part-time Lecturer** As a part-time lecturer at Waseda University's Graduate School of Fundamental Science and Engineering, I teach one class per week. My subject is C/C++ programming. At the same time, I am also a visiting researcher at Waseda University's research institute. My research interests include Transformer optimization and architecture design.
January 2023 **～Present**	**Online Webinar Lecture (Autonomous driving, TensorRT deployment)** I am coaching an online class in China similar to "Udemy", and offers the course related to "High Performance Implementation Using CUDA and TensorRT in Autonomous Driving". The content is deployment of various OSS SOTA models using CUDA acceleration methods and TensorRT API. In addition to these, it also includes computer architecture and parallel processing technology.
April 2022 **～Dec 2023**	**Honda Motor Co., Ltd., Electric Business Development Division, BEV Development Center, Software-Defund Mobility Development Department, Advanced Safety and Intelligent Solution Development Department, Intelligent Solution Laboratory, Engineer/Researcher** At Honda, I developed a multi-task DNN model for autonomous driving with tasks like detection, segmentation, keypoints, optical flow, and tracking. To improve performance on limited data, I applied techniques such as active learning and pseudo-labeling. I later began developing BEV models using Darknet and PyTorch. For real-time inference on resource-constrained in-vehicle hardware, I optimized the model with C++ parallelism, asynchronous execution, CUDA acceleration, and TensorRT layer-wise optimization. I also implemented custom CUDA ops, model quantization, pruning, and replaced low-efficiency operations in CNNs and Transformers to maximize performance. Skills: C/C++; CUDA & cuDNN & TensorRT; PyTorch; GO
April 2020 **~March 2021**	**Research Assistant, Graduate School of Information Science and Technology, Waseda University** During this period, I focused on the research that allows programs running on non-volatile memory to always maintain their correct program properties. I proposed a rational implementation method and formally proved that a program based on this implementation method can always operate correctly even in an uncertain execution environment, such as system crash. This goal is achieved by abstracting the memory structure of non-volatile memory, expressing the operation of memory propagation in a mathematical formula, and formally defining the grammar and semantics of the program. This research was published in a peer-reviewed academic journal. Skills: Python, RUST, C/C++, Hoare Logic, Rely-Guarantee programming, Memory consistency/persistency model, RISC-V, x86 low-level programming,
April 2019 **- September 2020**	**President, IEEE Eta-Kappa-Nu Mu-Tau Chapter** In September 2018, IEEE's world-class student academic organization called IEEE Eta-Kappa-Nu was founded as Japan's first IEEE student chapter (Mu-Tau Chapter) under Waseda University. I was appointed as president in 2019. While enrolled, I was involved in various activities in collaboration with student chapters in the United States and Europe, and won the Outstanding Chapter Award in Boston, USA in November of the same year. Skills: Leadership
Apr. 2017 **~March 2020**	**Waseda University, Faculty of Fundamental Science and Engineering, Department of Information Science and Technology, Research Associate** By using mathematics to examining whether a compiler is able to maintain the correct behavior and meaning of a program when it automatically parallelizes the program, it is possible to logically infer the execution of parallel programs. After abstractly defining the grammar and semantics of a program, the rationality of parallel program execution was formally defined, and the program could be verified. This research was carried out jointly with INRIA in France and the University of Arizona in the United States, and the contribution were successfully presented at international conferences and in journal. In addition to research, I also provides research guidance and experimental lectures for students. Skills: C/C++, NodeJS, Coq, Haskell, Compiler Optimization Technology, Parallel Processing Technology, Formal Verification, Model checking, Reasoning and Reasoning in Mathematical Theory, Logic Programming

Appealing points

Academic Disciplines

(A) Peer-reviewed journals, journals, and original papers

①(First)(Peer-reviewed) "Durable Queue Implementations built on a Formally Defined Strand Persistency Model", Journal of Information and Processing, Jun 2021, Jixin Han, Keiji Kimura

(B) Peer-reviewed papers at international conferences

①(First)(Peer-reviewed)"Parallelizing Compiler Translation Validation Using Happens-Before and Task-Set", The 9th International Workshop on Computer Systems and Architectures (CSA'21), Oct 2021, Jixin Han, Tomofumi Yuki, Michelle Mills Strout, Dan Umeda, Hironori Kasahara, Keiji Kimura.

②(Peer-reviewed)"Fast and Highly Optimizing Separate Compilation for Automatic Parallelization", The 2019 International Conference on High Performance Computing & Simulation (HPCS 2019), Jul 2019, Tohma Kawasumi, Ryota Tamura, Yuya Asada, Jixin Han, Hiroki Mikami, Keiji Kimura, Hironori Kasahara

③((Peer-reviewed)"Reducing parallelizing compilation time by removing redundant analysis", Proceedings of the 3rd International Workshop on Software Engineering for Parallel Systems, co-located with SPLASH 2016, Oct 2016, Jixin Han, Rina Fujino, Ryota Tamura, Mamoru Shimaoka, Hiroki Mikami, Moriyuki Takamura, Sachio Kamiya, Kazuhiko Suzuki, Takahiro Miyajima, Keiji Kimura, Hironori Kasahara

(C) Conference Presentations

①"Proving the Correctness of NVM Recovery under Strand Persistency Model", Information Processing Society of Japan (IPSJ-PRO), Sep 2021, Jixin Han, Keiji Kimura

②"The Translation validation for the Coarse-grain Parallelizing Compiler", Information Processing Society of Japan (IPSJ-PRO), Jixin Han, Tomofumi Yuki, Michelle Strout, Keiji Kimura

③"Verified Translation Validation Technique for OSCAR Automatically Parallelizing Compiler", Information Processing Society of Japan (IPSJ-PRO), Sep 2018, Jixin Han, Tomofumi Yuki, Michelle Strout, David Padua, Hironori Kasahara, Keiji Kimura

④"Code Generating Method with Profile Feedback for Reducing Compilation Time of Automatic Parallelizing Compiler", Information Processing Society of Japan, Special Interest Group on System Architecture (ARC217@ETNET2017), Mar. 2017, Rina Fujino, Jixin Han, Mamoru Shimaoka, Hiroki Mikami, Takahiro Miyajima, Moriyuki Takamura, Keiji Kimura, Hironori Kasahara

Awards & Qualifications

(1) IPSJ Yamashita Memorial Award (2023): "Durable Queue Implementations built on a Formally Defined Strand Persistency Model"Paper awarded

(2) IEEE HKN Outstanding Chapter Award (2020): IEEE HKN activities were recognized as IEEE HKN Japanese Chapter President.

(3) IEEE HKN Outstanding Chapter Award(2019):

(4) MCM/ICM (2014/2012): Participated in the Mathematical Modeling Contest and won the Honorable Mention in 2012 and the Meritorious Winner in 2014, respectively

(5) ACM/ICPC (2013): Participated in the ICPC contest when I was an undergraduate and eventually advanced to the Asia Regional Contest Beijing Area and won Silver

(6) Beijing Android development contest (2013): Participated in the Android development contest held in Beijing and won first place

(7) Language level: Japanese (Japanese Proficiency Test Level 1), English (TOEFL-iBT: 113/120), Chinese (native)

Programming experience

Programming Language (Mostly used): C/C++, python, CUDA

Programming Language (Moderate): GO, NodeJS

Programming Language (Occasionally Used): Rust, Haskell, Coq

Others: PyTorch, Darknet, TensorRT, LLVM